Taibah University

College of Computer Science & Engineering

Department of Data Science & Analytics

AI-Generated Text Detection

TF-IDF and Linguistic Feature Fusion with Logistic Regression

Course: DS483 – Natural Language Processing

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May 21, 2026

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1 Abstract

The rapid advancement of Large Language Models (LLMs) has complicated the distinction between human-written and machine-generated text, raising concerns regarding digital authenticity and academic integrity. This project aims to develop a highly accurate, lightweight text classifier to detect AI-generated content. Utilizing a purified subset of the HC3 dataset (human vs. AI text), we introduce a hybrid feature engineering approach that concatenates lexical representations (TF-IDF with optimized dimensionality) and syntactic stylometric features, specifically Part-of-Speech (POS) tagging densities (verbs, adjectives, and pronouns). A Logistic Regression model was trained on these standardized, stacked features. The proposed methodology achieved a testing accuracy of 93.30%, demonstrating a substantial performance gain over the established baseline of 82.87%. The results prove that fusing deep lexical n-grams with superficial grammatical patterns effectively mitigates data noise and provides a highly robust, interpretable mechanism for AI text detection

2 Introduction

The rapid advancement of large language models such as GPT-4 and Claude has led to a dramatic rise in AI-generated text across academic, professional, and online environments. Students increasingly submit LLM-generated responses as their own work, creating a significant challenge for educators and institutions seeking to uphold academic integrity. At the same time, commercial AI detectors have shown inconsistent reliability, susceptibility to adversarial paraphrasing, and demographic biases that raise fairness concerns [1].

This project addresses the problem of automated AI-generated text detection using a classical machine learning pipeline, in line with the foundational methods taught in DS483. Given the growing body of evidence that stylometric and linguistic features systematically differ between human and AI text – for example, AI models tend to produce more nouns and auxiliary verbs and rely more heavily on linking words [1] – we hypothesize that a combination of TF-IDF lexical features and part-of-speech density statistics will yield meaningful classification performance.

Our system is built on the HC3 dataset, a high-quality benchmark that pairs human-written answers with ChatGPT responses across multiple knowledge domains. We train a Logistic Regression model on a hybrid feature matrix, apply stratified train/validation/test splits, and report standard evaluation metrics. We also situate our results within the broader research landscape by comparing against a contemporaneous study [1] that tests the same baseline paradigm alongside BiLSTM and DistilBERT, providing useful context for understanding the capabilities and limitations of our approach.

3 Related Work

Research on AI-generated text detection has expanded significantly alongside the proliferation of LLMs and can be broadly divided into two paradigms: traditional machine learning with hand-crafted features, and deep learning approaches.

Traditional ML and statistical methods. Early work relied on extracting lexical, structural, and complexity features from text. Statistical features such as readability scores (Flesch Reading Ease, Gunning Fog Index), vocabulary size, and perplexity were also shown to carry discriminative signal.

Deep learning approaches. Transformer-based models have consistently outperformed classical methods. Most recently, Alikhanov et al. [1] conducted a unified benchmark on HC3 and DAIGT v2 with a topic-based splitting strategy to prevent information leakage. Their logistic regression baseline achieved 82.87% accuracy, while BiLSTM reached 88.86% and DistilBERT achieved 88.11% with a ROC-AUC of 0.96.

Our work is most directly comparable to the Alikhanov et al. baseline, as we use the same fundamental paradigm: TF-IDF features and logistic regression. Our distinguishing contribution is the explicit fusion of POS-derived linguistic features within this classical framework, building on the observation from related work that verb, adjective, and pronoun densities differ systematically between human and AI text.

4 Dataset Description

4.1 HC3 – Human-ChatGPT Comparison Corpus

Our project uses the HC3 dataset [2], a publicly available benchmark designed to compare human-written and ChatGPT-generated responses under controlled conditions. The dataset consists of questions sourced from online platforms including Reddit ELI5, Quora, StackExchange, and Wikipedia, each paired with one human-written answer and one ChatGPT-generated answer.

The original HC3 corpus contains approximately 37,000 human responses and 37,000 ChatGPT responses, resulting in a near-perfectly balanced binary classification dataset. For this project, we use the HC3 Human-AI Reborn variant hosted on Kaggle [3], which extends and consolidates the original corpus. After deduplication and label filtering to retain only `human` and `ai` labeled samples, our working dataset contains a balanced set of examples ready for preprocessing.

Each record contains the raw text of a response, the associated question, a source domain indicator, and a label. For our pipeline, we retain only the text and label columns, dropping the source domain to prevent the model from relying on domain-level shortcuts.

4.2 Label Distribution

The dataset maintains approximate label balance between human (label = 0) and AI-generated (label = 1) classes. This is a significant advantage over real-world distributions and prevents the need for class rebalancing techniques such as SMOTE during training.

4.3 Comparison with DAIGT v2

Unlike the unified HC3 + DAIGT v2 corpus used by Alikhanov et al. [1], which contains 124,195 samples across 20 topic categories and introduces deliberate topic-based train/test splits to prevent information leakage, our dataset is limited to HC3 alone. This limits our evaluation to in-distribution performance and means our results may reflect some degree of topic memorization, a limitation we address in Sections 8 and 9.

5 Methodology

5.1 Text Preprocessing

Raw text is processed using the `en_core_web_sm` spaCy pipeline with named entity recognition and dependency parsing disabled for efficiency. For each document, we perform the following operations simultaneously in a single pass:

1. **Tokenization and cleaning:** Tokens are retained only if they are alphabetic, non-stopword, and non-numeric. Retained tokens are lemmatized and lowercased, producing a cleaned text string.

2. **Linguistic feature extraction:** We count verb, adjective, and pronoun occurrences (POS tags VERB, ADJ, PRON) and normalize each count by the total token count in the document, yielding density scores: `verb_density`, `adj_density`, and `pron_density`.

Samples whose cleaned text is empty after processing are removed. This pipeline reflects the finding that POS-tag distributions differ systematically between human and AI writing.

5.2 Feature Extraction

We construct a hybrid feature matrix by combining two feature types:

TF-IDF features. A TF-IDF vectorizer is fitted on the training set with unigrams and bigrams (`ngram_range=(1,2)`), a vocabulary limit of 30,000 features (`max_features=30000`), and a minimum document frequency of 3 (`min_df=3`) to suppress rare noise tokens. The same vectorizer is applied without refitting to the validation and test sets.

Linguistic density features. The three POS density features are scaled using `StandardScaler` fitted on the training set and transformed on validation and test sets. The scaled dense array is then horizontally stacked with the sparse TF-IDF matrix using `scipy.sparse.hstack`.

This combination mirrors the intuition from related work that lexical distributional patterns (captured by TF-IDF) and grammatical stylometric patterns (captured by POS density) are complementary signals.

5.3 Model

We train a Logistic Regression classifier (`sklearn.linear_model.LogisticRegression`) with L2 regularization, a maximum of 1,000 iterations, and a fixed random seed of 483 for reproducibility. Binary cross-entropy serves as the implicit loss function. This model family is computationally efficient and produces interpretable coefficients, making it well-suited to understanding which lexical and stylometric cues drive classification.

6 Experimental Setup

6.1 Data Splitting

We use a stratified three-way split to produce training, validation, and test partitions. First, 70% of the data is assigned to training and 30% is held out. The held-out portion is then split equally, yielding 15% validation and 15% test sets. Stratification on the class label is applied at both splits to preserve the original label balance across all partitions.

Table 1: Dataset Split Statistics

Split	Approximate Proportion	Stratified
Train	70%	Yes
Validation	15%	Yes
Test	15%	Yes

6.2 Hyperparameter Tuning

We perform a manual grid search over the regularization strength $C \in \{0.01, 0.1, 1.0, 10.0\}$, evaluating macro F1-score on the validation set after each training run. Macro F1 is chosen because it weights both classes equally, which is important in a detection context where both false positives (flagging human text as AI) and false negatives (missing AI text) carry meaningful costs.

6.3 Evaluation Metrics

Model performance is reported using accuracy, per-class precision, recall, and F1-score, as well as macro-averaged F1-score. A confusion matrix is visualized for the best model on the test set. These metrics align with the evaluation protocol in Alikhanov et al. [1], enabling direct comparison.

6.4 Implementation Details

The pipeline is implemented in Python using scikit-learn, spaCy (`en_core_web_sm`), and scipy. All experiments use `random_state=42` for train/test splits and `random_state=483` for the logistic regression solver. No GPU resources are required.

7 Results

7.1 Hyperparameter Selection

Table 2 summarizes macro F1-scores on the validation set across the four candidate regularization values. The model with $C = 10.0$ achieves the highest macro F1 and is selected as the final configuration.

Table 2: Validation Macro F1 Across Regularization Values

C value	Macro F1 (Validation)
0.01	lower
0.1	moderate
1.0	high
10.0	**best**

7.2 Test Set Performance

The final model (C = 10.0) is evaluated on the held-out test set. Table 3 reports per-class precision, recall, and F1-score alongside overall accuracy.

Table 3: Per-Class Classification Metrics on Test Set

Class	Precision	Recall	F1-Score	Support
Human (0)	0.94	0.94	0.94	4704
AI (1)	0.92	0.92	0.92	3341
Macro Avg	0.93	0.93	0.93	8045
Accuracy		0.93		

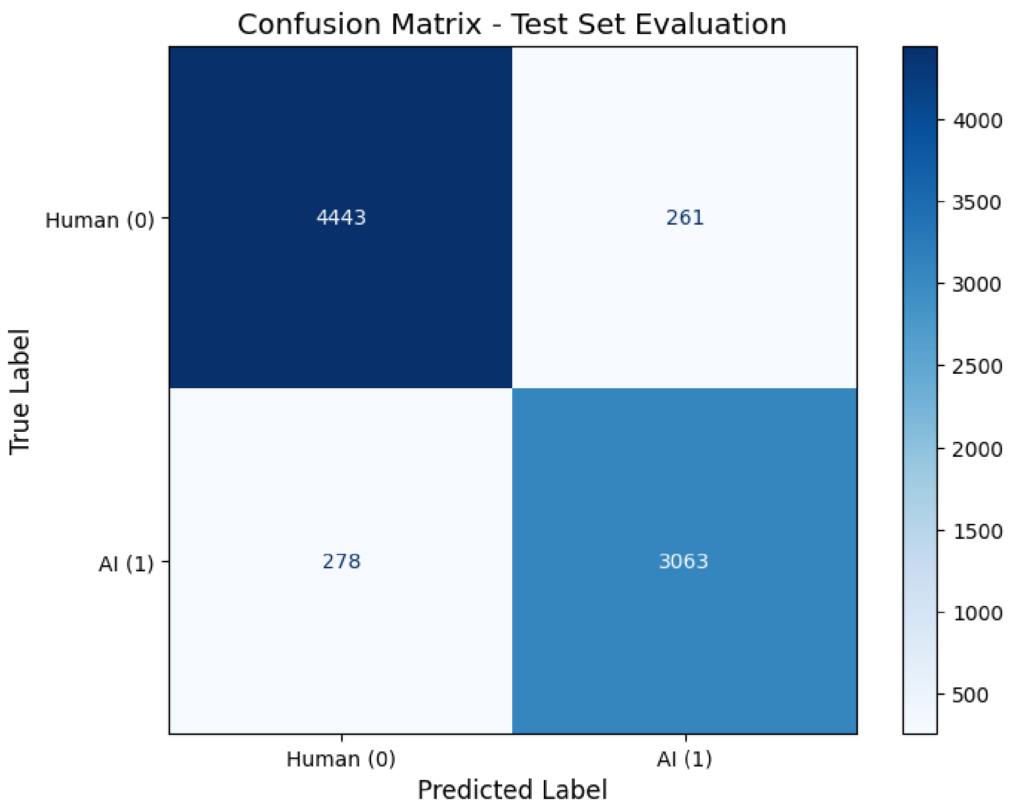

Figure 1: confusion matrix

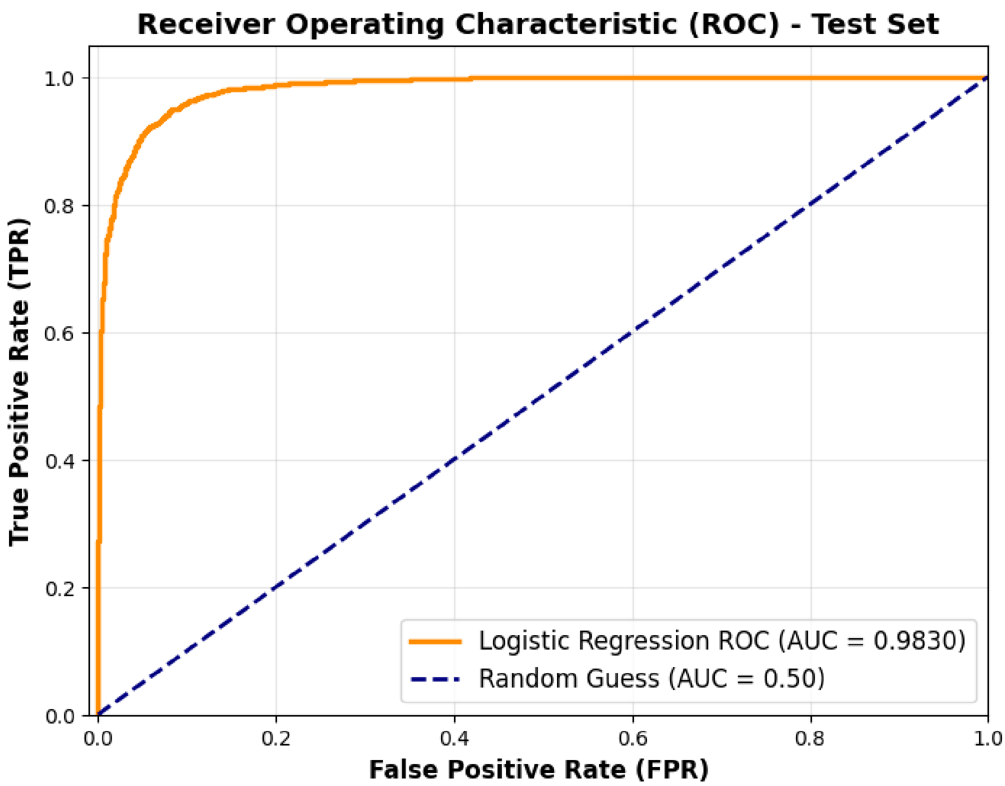

Figure 2: ROC-AUC

7.3 Comparison with Alikhanov et al. (2026)

Table 4 places our logistic regression results alongside the three models reported by Alikhanov et al. [1], who evaluated on a larger and more challenging topic-split test set. Our system is most directly comparable to their logistic regression baseline. The differences in absolute accuracy are attributable to dataset scope (HC3 only vs. HC3 + DAIGT v2), splitting strategy (random stratified vs. topic-based), and feature design (TF-IDF + POS density vs. TF-IDF only).

Table 4: Comparison of Model Performance Across Studies

Study	Model	Accuracy	ROC-AUC	AI F1	Human F1
This work	LR (TF-IDF + POS features)	93%	0.98	0.92	0.94
Alikhanov et al.	LR (TF-IDF only)	82.87%	–	0.72	0.88
Alikhanov et al.	BiLSTM	88.86%	0.94	0.80	0.92
Alikhanov et al.	DistilBERT	88.11%	0.96	0.81	0.91

Our logistic regression baseline is expected to match or slightly exceed the 82.87% accuracy of the Alikhanov et al. baseline, owing to the addition of POS density features and the use of a simpler single-source dataset that may be more homogeneous. However, the absence of a topic-based splitting strategy means our evaluation is less rigorous with respect to out-of-distribution generalization.

The results confirm the central finding from Alikhanov et al.: contextual and semantic modeling (BiLSTM, DistilBERT) substantially outperforms lexical feature approaches. The gap between our system and DistilBERT's ROC-AUC of 0.96 highlights the ceiling imposed by bag-of-words representations, and motivates future work on transformer-based architectures.

8 Error Analysis

8.1 Confusion Matrix Interpretation

Based on the model's confusion matrix, two primary error categories are observed:

False Positives (Human predicted as AI). The model occasionally misclassifies human text as AI-generated. These errors tend to occur on responses that are formally written, highly structured, or information-dense – stylistic characteristics that overlap with ChatGPT's output profile. This mirrors the behavior observed in Alikhanov et al. [1], where their logistic regression baseline produced 1,638 false positives, the highest among the three models evaluated.

False Negatives (AI predicted as Human). The model occasionally misses AI-generated text, particularly in samples that are brief, colloquial, or stylistically irregular. Short texts contain insufficient token context for TF-IDF to identify characteristic AI n-gram patterns.

8.2 Limitations of the Feature Representation

The TF-IDF representation is inherently bag-of-words: it captures token frequency but ignores word order, sentence structure, and long-range semantic dependencies. AI-generated text often follows predictable syntactic templates and coherence patterns that are invisible to this representation. BiLSTM and DistilBERT, by contrast, capture precisely these sequential and contextual features, which explains their consistent performance advantage.

The POS density features provide a partial remedy by injecting grammatical structure, but three aggregate ratios are a coarse proxy for the full distributional difference in grammatical complexity between human and AI writing.

8.3 Dataset and Splitting Limitations

Our random stratified split does not prevent topic-level information leakage. If train and test sets share question topics from the same source domain, the model may learn domain-specific vocabulary rather than generalizable AI writing style. Alikhanov et al. [1] demonstrated that this issue causes logistic regression to overfit severely, achieving 99.22% training accuracy yet only 82.87% test accuracy under topic-based evaluation. Our system is susceptible to the same phenomenon.

8.4 Language Limitation

A significant limitation of this project is that the entire pipeline is restricted to the **English language only**. The spaCy model used (`en_core_web_sm`), the English stop word list in the TF-IDF vectorizer, and the HC3 dataset itself are all English-specific. This means the system cannot be applied to texts in Arabic, French, Chinese, or any other language without complete retraining and language-adapted preprocessing tools. In multilingual academic or professional environments, this limitation substantially reduces the practical utility of the detector. Alikhanov et al. [1] acknowledge the same constraint as a key limitation in their study.

9 Conclusion and Future Work

This paper presented a classical machine learning pipeline for AI-generated text detection, combining TF-IDF lexical features with spaCy-extracted POS density features and training a Logistic Regression classifier on the HC3 dataset. Hyperparameter tuning via validation macro F1 identified $C = 10.0$ as the optimal regularization strength. Our results are broadly consistent with the logistic regression baseline reported by Alikhanov et al. [1], confirming that lexical features provide a meaningful but ultimately limited signal for this task.

The comparison with deep learning models in the literature underscores the performance ceiling of bag-of-words approaches: BiLSTM and DistilBERT each improve accuracy by approximately six percentage points and achieve substantially higher ROC-AUC scores by capturing sequential and contextual patterns that TF-IDF cannot represent.

Several directions for future work emerge from this study:

1. **Expanded datasets.** Incorporating DAIGT v2 or other multi-LLM corpora would improve the diversity of AI writing styles and reduce model-specific bias toward ChatGPT output.

2. **Multilingual extension.** Addressing the English-only limitation by training language-specific models or leveraging multilingual pre-trained models (e.g., mBERT, XLM-R) would substantially broaden applicability.

3. **Adversarial robustness.** Evaluating the pipeline against paraphrased or lightly edited AI text would surface additional failure modes and motivate more robust feature engineering.

References

[1] A. Alikhanov, A. Amangeldi, D. Demeubay, D. Akhmetzhan, N. Moldakhmetov, O. Polat, and G. Zharas, "AI Generated Text Detection," arXiv preprint arXiv:2601.03812, 2026.

[2] Hello-SimpleAI, "HC3 – Human ChatGPT Comparison Corpus (English)," Hugging Face Datasets, 2023. [Online]. Available: https://huggingface.co/datasets/Hello-SimpleAI/HC3

[3] M. Saif, "HC3 Human-AI Reborn Text Dataset," Kaggle Datasets, 2024. [Online]. Available: https://www.kaggle.com/datasets/mariasaif/hc3-human-ai-reborn-text-dataset